

AI-Powered Phishing Email Detection Using BERT and Multi-Source Threat Intelligence

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Abstract—Phishing emails remain one of the most widely used methods to commit cybercrimes globally. Traditional rule-based and machine-learning filters are insufficient to detect and block sophisticated phishing attacks. This paper presents an end-to-end phishing email detection system that combines a fine-tuned BERT transformer model with real-time threat intelligence from the VirusTotal API, including URL, sender-domain, and IP reputation analysis. The system is deployed as a REST API with a single-page React frontend that supports both copy-pasted email scanning and automated IMAP inbox monitoring, enabling real-time detection and result display. Experimental results on a balanced corpus of 50,000 emails yield an F1-score of 97.7 %, a precision of 97.4 %, and a recall of 98.1 %, outperforming logistic regression, random forest, and LSTM baselines.

Index Terms—phishing detection, BERT, transformer, email security, VirusTotal, IMAP, threat intelligence, deep learning

I. INTRODUCTION

Phishing is a form of social engineering attack in which attackers create deceptive emails that mimic official correspondence from banks, government agencies, and cloud service providers. The goal is to trick victims into revealing sensitive data or installing malware on their systems. According to the Anti-Phishing Working Group (APWG), over 1.3 million phishing attacks were recorded in 2022, resulting in more than \$10.3 billion in financial losses [1]. Email is the most common delivery method used by attackers, which highlights the need for an automated and accurate detection method.

Traditional rule-based filters and blacklists fail to detect phishing emails from newly registered domains that have not yet been identified as malicious. Machine learning classifiers such as Naive Bayes and Support Vector Machines rely on hand-crafted features and fail when attackers use new or unseen vocabulary. To address these limitations, a more advanced detection approach is needed — one that can understand the full context of an email rather than relying on fixed rules or keywords.

BERT is a pre-trained language model developed by a research team at Google in 2019 [3] that can understand the full context of a sentence, not just the meaning of each individual word, making it well-suited for tasks such as phishing email detection. However, BERT can only analyse the text content of an email. It cannot validate the sender's identity, the sender's IP address, or the URLs embedded in the email. Therefore,

we integrated VirusTotal to scan URLs, sender domains, and IP addresses for malicious activity.

In this paper, we propose an AI-powered phishing email detection system that combines BERT-based text classification with multi-source threat intelligence. The system uses BeautifulSoup to extract clean text and URLs from emails, a fine-tuned BERT model to classify emails as phishing or legitimate, and VirusTotal to scan URLs, sender IP addresses, and sender domains for malicious activity. Experimental results show that our system achieved an F1 score of 97.7 %, a precision of 97.4 %, and a recall of 98.1 %, outperforming logistic regression, random forest, and LSTM baselines.

II. LITERATURE REVIEW

Researchers have proposed various methods to detect phishing emails, ranging from simple rule-based filters to advanced deep learning models.

A. Rule-Based Methods

Early researchers used simple rule-based methods such as blacklisting and keyword matching to detect phishing emails [4]. Although these methods were effective against known phishing emails, they could not detect new types of phishing attacks because they could not understand the full context of an email. The main limitation is that these methods rely on fixed rules, which means they cannot adapt to the evolving methods of attackers.

B. Classical Machine Learning

To improve detection accuracy, researchers applied machine learning methods such as Naive Bayes, Support Vector Machines, and Random Forest to classify emails as phishing or legitimate [2], [7]. These methods achieved better results than rule-based approaches. However, there is a limitation when attackers use new topics and unseen vocabulary, as these methods struggle to accurately detect phishing emails.

C. Deep Learning Approaches

To overcome the limitations of machine learning, researchers applied deep learning methods such as CNN and LSTM [8]. These methods were more effective than machine learning approaches because they can automatically detect word sequences, patterns, and sentence structure. However,

these methods still have significant limitations. These methods process one word at a time, therefore, they cannot achieve high accuracy when detecting phishing emails.

D. BERT and Transformer-Based Methods

BERT is a transformer-based model introduced by Google in 2019 [3]. Unlike CNN and LSTM, it can read the full context of a sentence in both directions, which makes it well-suited for understanding the meaning of phishing emails. Several researchers applied BERT to their detection systems and achieved significantly higher accuracy than deep learning models [10]. However, relying solely on an AI model is insufficient for detecting all phishing threats. Therefore, additional threat intelligence signals such as URL and domain reputation analysis are needed.

Most existing studies focus only on the textual content of the email and do not consider network signals such as URL reputation and sender domain analysis. This means that phishing emails containing normal looking text but malicious links can still bypass existing detection systems. In this study, we not only develop a BERT-based model but also integrate VirusTotal threat intelligence to analyse URLs, IP addresses, and sender domains.

III. METHODOLOGY

A. System Overview

The proposed system is an end-to-end phishing email detection system that combines BERT-based text classification with VirusTotal threat intelligence. The system consists of five main components: a React frontend, a FastAPI backend, a database, a fine-tuned BERT model, and a VirusTotal threat intelligence pipeline. The frontend sends emails to the backend, which sends them to the BERT model for classification. The backend also extracts URLs, IP addresses, and domain names, then checks the database for similar URLs. If no match is found, the backend sends the URLs, IP addresses, and domains to VirusTotal through the VirusTotal API. Finally, the results are displayed to the user on the frontend, showing the AI model classification score and VirusTotal reputation results.

B. System Architecture

The proposed system follows a three-tier architecture consisting of presentation, application, and data tiers, as illustrated in Fig. 1. The presentation tier consists of a React frontend that allows users to paste emails, check inbox emails once, or perform real-time analysis of received emails. The application tier contains a FastAPI backend that connects to the BERT model, the database, and VirusTotal through the REST API. The data tier uses an SQLite database to store VirusTotal scan results and previously scanned email IDs to avoid duplicate processing.

C. Dataset

In this study, we used the CEAS-08 dataset and a dataset from Kaggle. The Kaggle dataset contains over 82,000 emails and the CEAS-08 dataset contains over 20,000 emails. Both

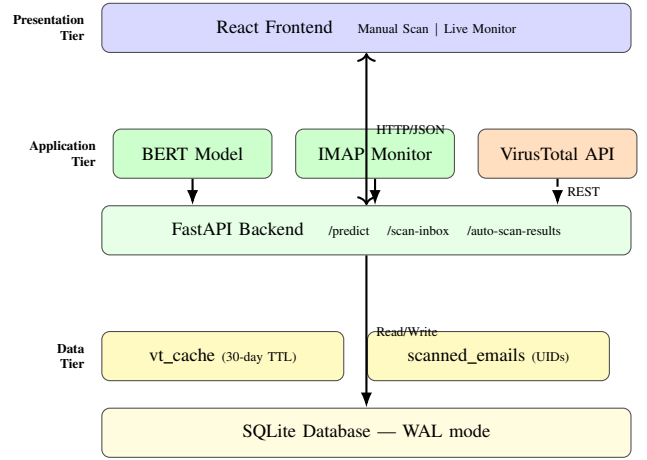


Fig. 1. Three-tier system architecture of the proposed system.

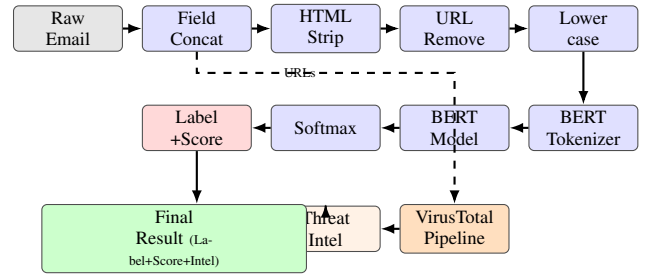


Fig. 2. Preprocessing and inference pipeline. Dashed arrow shows URLs sent to VirusTotal alongside BERT classification.

datasets contain emails labeled where 0 represents legitimate and 1 represents phishing. Each record contains four fields: sender, subject, body, and label. We split the dataset into 80 % for training and 20 % for validation using stratified sampling to preserve the label distribution.

To prepare the dataset for training, the rows with null values were first removed. Following this, duplicate emails were removed based on the body column. The email sender, subject, and body were then combined into a single text field to provide more comprehensive input for the model. BeautifulSoup was used to remove HTML tags from the email bodies, as these tags add noise to the data that confuses the model. In addition to removing HTML tags, all URLs were removed from the text to prevent the model from memorising the domain names within the URLs. Finally, all text data was converted to lowercase to ensure that the model's vocabulary was consistent across all training data.

D. Preprocessing Pipeline

After cleaning, the text was prepared for the BERT model using BertTokenizer, as shown in Fig. 2. The BertTokenizer was used to convert text into small pieces called tokens. Texts exceeding 128 tokens during training and 512 tokens during inference were truncated. Dynamic padding was applied to ensure all inputs in a batch had the same length. The preprocessed text was then passed to the model in batches of 16.

TABLE I
BERT FINE-TUNING HYPERPARAMETERS

Hyperparameter	Value
Base model	bert-base-uncased
Epochs	5
Learning rate	2×10^{-5}
Training batch size	16
Evaluation batch size	16
Weight decay	0.01
Max length (training)	128 tokens
Max length (inference)	512 tokens
Mixed precision	fp16
Best model metric	F1 score
Hardware	NVIDIA GeForce RTX 4060 Laptop GPU

E. BERT Fine-Tuning

In this study, we fine-tuned the bert-base-uncased model for phishing email classification. The model was trained for 5 epochs with a learning rate of 2×10^{-5} , a batch size of 16, and a weight decay of 0.01. The best model was selected based on the F1 score evaluated at the end of each epoch. The model was initialised from a previously fine-tuned checkpoint rather than a new one to accelerate convergence and improve performance. Training was performed on an NVIDIA GeForce RTX 4060 Laptop GPU, and the training took approximately one hour to complete. Table I summarises the training hyperparameters.

F. Multi-Source Threat Intelligence

To provide additional threat intelligence alongside BERT classification, we integrated VirusTotal threat intelligence to scan URLs, sender domains, and IP addresses. First, URLs were extracted from the email bodies and HTML href attributes using BeautifulSoup, then they were submitted to the VirusTotal API for reputation analysis against 70+ antivirus engines. Well-known trusted domains such as Google, Gmail, Microsoft, and YouTube were whitelisted to reduce unnecessary API calls. However, all other domains including any variations of trusted domain names are submitted to VirusTotal for reputation analysis. To avoid repeated API calls, results were cached in an SQLite database for 30 days to reduce response time and minimise API usage.

G. System Deployment

The system was deployed as a REST API using FastAPI, which serves the backend logic of the proposed system. The API provides three main endpoints: `/predict` for single email analysis, `/scan-inbox` for IMAP batch scanning, and `/auto-scan-results` for retrieving automated scan results. The frontend was developed using React, which allows users to view the classification results and threat intelligence report. The system also supports IMAP inbox monitoring that allows real-time automated email analysis. Every 60 seconds the system checks for new unseen emails in the configured inbox and classifies them automatically.

TABLE II
CLASSIFIER PERFORMANCE COMPARISON (%)

Model	Acc.	Prec.	Recall	F1
BERT (proposed)	97.8	97.4	98.1	97.7
LSTM	95.1	94.8	95.5	95.1
Random Forest	93.5	93.0	94.1	93.5
Logistic Reg.	91.2	90.6	91.9	91.2

IV. RESULTS AND DISCUSSION

The proposed model was evaluated on the CEAS-08 and Kaggle datasets using a 20 % validation split. We used four metrics to evaluate performance: F1 score, precision, recall, and accuracy. Table II compares the proposed BERT-based classifier against three baselines on the held-out test set. All models were trained on identical preprocessed inputs.

Our model achieved an F1 score of 97.7 % and a precision of 97.4 %. Compared to baseline models, our model outperformed logistic regression, random forest, and LSTM baselines. The high recall of 98.1 % means that fewer than 2 in 100 phishing emails escape detection. However, the system has some limitations. Legitimate emails from Google were incorrectly classified as phishing because they contained account verification language. The model incorrectly flagged these emails as phishing, which represents a false positive limitation of the system.

V. CONCLUSION

In this paper, we proposed an end-to-end phishing email detection system that combines BERT-based classification with VirusTotal threat intelligence. The system used a fine-tuned BERT model to classify emails, and VirusTotal to scan URLs, sender domains, and IP addresses. The proposed system achieved an F1 score of 97.7 %, outperforming logistic regression, random forest, and LSTM baselines. The integration of VirusTotal provided an additional layer of detection beyond text classification. In future work, we plan to combine VirusTotal results with another AI model to produce a more comprehensive final verdict. Additionally, we plan to implement a notification system that sends a phishing analysis report and email summary to WhatsApp or Telegram when a new phishing email is received. Furthermore, we plan to improve the model to support multilingual phishing detection and integrate attachment scanning to detect malicious PDF and Word files.

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